

Requirements Specification Document

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Introduction

Purpose: To create a children’s toy designed to teach young children the alphabet, encourage critical thinking, and enhance their vocabulary.

Scope:

Justification	To design and create a unique and interactive toy that focuses on supplying educational information as well as incorporating critical thinking questions
Description	• Will be able to instruct about the letters of the alphabet and words that begin with them • Will be able to provide a service that offers critical thinking
In Scope	• Learning mode • Quiz mode • Holographic display • Rotating case • 26 alphabetical buttons • 2 buttons to change mode
Out of Scope	• More game modes • Additional screens • Additional buttons

System Overview: The system will be a children’s toy that resembles a stereotypical alien spaceship or UFO.

The software of the toy will implement a “learning mode” for teaching the alphabet. In this mode, the child will press a button corresponding to a letter of the alphabet, and the toy will provide audiovisual feedback of the letter pressed.

The “quiz mode” of the toy will be aimed at children building a vocabulary. It will project holographic images and prompt a child to guess the first letter of the word corresponding to what the image portrays. Upon receiving input, the toy will provide audiovisual feedback on whether the input was correct or not. The hardware will draw images randomly from a word bank, and implement a record of correct/incorrect answers for a game. It will also keep a record of correct/incorrect guesses for each word in the word bank and use this as a dynamic measure of difficulty.

Finally, the system will also implement installation of different word banks. These can be of increasing difficulty as a child’s vocabulary grows, or tailored to different subjects.

Definitions

- A database of letters will be constructed for Learning Mode that contains an image and a sound of the letter being pronounced.
- A database of word banks or quizzes will be constructed. These will be grouped by difficulty or reading level.

Use Cases

Use Case: UC1	Power On
Primary Actor	User
Secondary Actor	Toy
Precondition	• Toy must be plugged into a power supply • User must push the power button
Postcondition	• Toy must power on • Toy must generate some image • Toy must output some audio

Use Case: UC2	Power Off
Primary Actor	User
Secondary Actor	Toy
Precondition	• Toy must be plugged into a power supply • Toy must be powered on
Postcondition	• Power supply to toy must disconnect

Use Case: UC3	Learning Mode
Primary Actor	User
Secondary Actor	Toy
Precondition	• Toy must be powered on • Learning Mode is the default mode
Postcondition	• Stay in Learning Mode unless there is no power supply • Stay in Learning Mode unless Quiz Mode has been selected

Use Case: UC4	Quiz Mode
Primary Actor	User
Secondary Actor	Toy
Precondition	• Toy must be powered on • Quiz Mode button must be pushed
Postcondition	• Stay in Quiz Mode unless there is no power supply • Stay in Quiz Mode unless Learning Mode is selected • Return to the menu after a Quiz mode game finishes

Functional Requirements

Software Requirements

- i. The system will have two different modes: “Learning” and “Quiz”
- ii. Learning Mode will allow the user to press each alphabet button and listen to how the letter sounds.
- iii. Quiz Mode will allow the user to see holographic images and guess which letter they start with.
- iv. Software must correlate an image to a word (quiz mode)
- v. Software must correlate audio to a word (quiz mode)
- vi. Software must draw random entries from a wordbank (quiz mode)
- vii. Software must accept an input from the user (quiz mode and learning mode)
- viii. Software must return a response based on validity of the input (quiz mode)
- ix. Software must be able to implement addition of new and different wordbanks (quiz mode)

Hardware Requirements

- i. Case and buttons must be able to rotate around base
- ii. System must be able to turn off/on
- iii. System must contain 26 alphabet buttons
- iv. System must display holographic image
- v. System must be able to output audio

Nonfunctional Requirements

Software Requirements

- i. Should be responsive
- ii. Should be simple enough for a child to turn on and start the desired game mode
- iii. Should be stable

Hardware Requirements

- i. Buttons should be easy for a small child to press
- ii. The toy should be safe for a small child to use
 - a. No choking hazards
 - b. Not so heavy that it can injure a small child
 - c. Shouldn't be an electrocution hazard for small children
- iii. Toy shouldn't be too fragile for a small child to use
- iv. Battery life should be powerful enough to safely operate toy

Change Control Board

- The requester of a project change must present an overview of said change(s) along with details about the effects of the change
- The request will be presented to the team and be reviewed as a team
- The change will only occur after majority vote with full understanding of said change(s)

References:

- <https://www.youtube.com/watch?v=BSGDJeI2vEU#action=share>
- <https://www.youtube.com/watch?v=9t0cOYvOy4M>